Chapter: 4

Redox Reactions

1) What is the process of losing electrons in a chemical reaction called?
a) Reduction b) Oxidation c) Ionization d) Neutralization

2) Which gas is released when zinc reacts with hydrochloric acid (HCl)?
- a) Oxygen b) Nitrogen c) Hydrogen d) Carbon dioxide

3) What law states that the total mass of reactants is equal to the total mass of products in a chemical reaction?
- a) Law of Conservation of Energy b) Law of Conservation of Mass
- c) Law of Conservation of Momentum d) Law of Definite Proportions

4) What is the oxidation number of oxygen in most of its compounds?
- a) +1 b) +2 c) -1 d) -2

5) In the reaction $2H_2 + O_2 \rightarrow 2H_2O$, which element undergoes reduction?
- a) Hydrogen b) Oxygen c) Both hydrogen and oxygen d) Neither hydrogen nor oxygen

6) What is the term for a substance that undergoes reduction in a chemical reaction?

7) What is the oxidation number of chlorine in NaCl?

8) What element is reduced in the reaction $Zn + 2HCl \rightarrow ZnCl_2 + H_2$?

9) In the reaction $4Fe + 3O_2 \rightarrow 2Fe_2O_3$, what happens to iron (Fe)?
- a) It is reduced. b) It is oxidized. c) It is neutralized. d) It remains unchanged.

10) What is the reducing agent in the reaction $2Na + Cl_2 \rightarrow 2NaCl$?
- a) Sodium b) Chlorine c) Sodium chloride d) Water

11) Which of the following reactions involves both oxidation and reduction?
- a) Decomposition reaction b) Combustion reaction c) Redox reaction d) Acid-base reaction

12) In the reaction $Zn + 2HCl \rightarrow ZnCl_2 + H_2$, what is the role of Zn?
- a) Reducing agent b) Oxidizing agent c) Catalyst d) Product

13) Which element in the reaction $2H_2 + O_2 \rightarrow 2H_2O$ undergoes oxidation?
- a) Oxygen b) Hydrogen c) Water d) None

14) What is the product when methane (CH_4) burns in oxygen?
- a) Carbon dioxide and water b) Carbon monoxide and hydrogen c) Methanol and oxygen d) Carbon dioxide and oxygen

15) Which substance is reduced in the reaction $Mg + F_2 \rightarrow MgF_2$?
- a) Magnesium b) Fluorine c) Both magnesium and fluorine d) Neither magnesium nor fluorine

16) What term is used for a reaction in which oxidation and reduction occur simultaneously?

17) What is the oxidation number of sodium (Na) in NaCl?

18) In the reaction $H_2 + Cl_2 \rightarrow 2HCl$, which element undergoes oxidation?

19) Which gas is produced during the combustion of methane (CH_4)?

20) What is the oxidation number of oxygen in H_2O?

21) Which ion is formed when magnesium loses two electrons?

22) Which element has an oxidation number of +3 in aluminium oxide (Al_2O_3)?

23) Given the reaction: $2Mg + O_2 \rightarrow 2MgO$
Explain how magnesium is oxidized and identify the oxidizing agent.

24) Balance the chemical equation for the combustion of methane and verify if it obeys the law of conservation of mass:
$CH_4 + O_2 \rightarrow CO_2 + H_2O$

25) In the reaction: $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$, identify the substance that is reduced and the substance that is oxidized. Explain why.

26) What is a redox reaction? Give an example.

27) Explain how the law of conservation of mass is applied to the reaction:
$2H_2 + O_2 \rightarrow 2H_2O$

28) Explain the role of an oxidizing agent in a redox reaction.

29) What are the products formed when sodium reacts with water? Write the balanced chemical equation.

30) Why is the total mass of the products equal to the total mass of the reactants in a chemical reaction? Illustrate with an example.

31) Assertion (A): Sodium (Na) loses an electron to form a positive ion during the formation of sodium chloride (NaCl).
Reason (R): Sodium undergoes oxidation by gaining electrons.

- a) Both A and R are true, and R is the correct explanation of A.
- b) Both A and R are true, but R is not the correct explanation of A.
- c) A is true, but R is false.
- d) A is false, but R is true.

32) Assertion (A): In a redox reaction, the substance that gets reduced is the reducing agent.
Reason (R): The reducing agent undergoes oxidation during a redox reaction.

- a) Both A and R are true, and R is the correct explanation of A.
- b) Both A and R are true, but R is not the correct explanation of A.
- c) A is true, but R is false.
- d) A is false, but R is true.

33) Assertion (A): The total mass of the reactants is equal to the total mass of the products in a chemical reaction.
Reason (R): Antoine Lavoisier proposed the law of conservation of mass.

- a) Both A and R are true, and R is the correct explanation of A.
- b) Both A and R are true, but R is not the correct explanation of A.
- c) A is true, but R is false.
- d) A is false, but R is true.

34) Atoms participate in chemical bonding by losing, gaining or sharing ----------
a) Electrons
b) Protons
c) Nuclei
d) Neutrons

35) The process of involving loss of electrons in a chemical reaction is -------
a) Oxidation
b) reduction
c) Dehydration
d) Combustion

36) Generally, metals undergo --------- during chemical reactions.
a) accepts electrons. b) reduction takes place.
c) Oxidation takes place. d) becomes diatomic

37) $2Na + Cl_2 \rightarrow 2NaCl$
Which is the oxidising agent in this chemical reaction?
a) Na b) Cl_2 c) NaCl d) None of these.